

PARTHIV RAWAT

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SUMMARY

Full-Stack / Backend Engineer with 3+ years of experience building low-latency distributed systems in Go and cloud-native applications on GCP and AWS. Specialized in backend architecture, caching, microservices, and AI-enabled platforms, with proven success improving latency by up to 45%, scaling services to 10K+ daily transactions, and delivering highly reliable production systems.

EXPERIENCE

MethodWorks Pvt. Ltd.

May 2023 – Present

Software Engineer

Noida, India

- Designed and built Golang REST APIs with sub-100ms P95 latency under high-concurrency workloads validated through load testing.
- Scaled 40+ backend APIs to handle 10K+ daily transactions with 99.9% uptime on GCP App Engine, with auto-scaling configuration handling 5x traffic spikes.
- Architected multi-layer Redis caching strategy using write-through and cache-aside patterns across 15+ services, reducing average API response time from 180ms to 99ms and database load from 850 req/s to 425 req/s.
- Redesigned indexing strategy and query execution paths across MySQL and MongoDB, reducing P95 response latency from 420ms to 270ms.
- Implemented micro-frontend architecture using Nx Module Federation with shared dependencies and lazy loading, reducing release cycles by 40% and bundle size by 25%.
- Built enterprise admin console with role-based access control (RBAC) for 500+ admin users, improving monitoring efficiency by 60% through real-time dashboards and alerting.
- Designed JWT-based authentication and authorization service with role-based access control, secure credential management using bcrypt hashing, and session refresh token rotation.
- Implemented event-driven architecture using message queues for asynchronous processing, improving system resilience and enabling horizontal scaling.
- Designed circuit breaker pattern implementation for external service dependencies, reducing cascade failures and improving system availability during outages.

Gao Tek Inc.

(Jan 2023 - Apr 2023)

Software Development Intern, Internship

Bangalore, India

- Debugged and optimized legacy codebase, fixing 15+ critical bugs and improving system stability by 30% through code modularization and error handling improvements.
- Integrated APIs, hardware drivers, and modem interfaces into unified library using adapter pattern, improving maintainability and reducing integration complexity by 40%.

PROJECTS

AI Search & Conversational Platform

Full-Stack AI Platform

- Architected full-stack **AI-powered search platform** with React frontend, Go backend, and SSE streaming for real-time conversational AI responses with memory-aware context.
- Built personalized search engine with trie-based autosuggestions, vector embeddings for semantic search, and Redis caching layer for geolocation-aware results.
- Implemented AI agent framework with tool invocation system, RAG pipelines, and real-time web search integration using REST APIs and JSON-RPC communication.

Educational Technology Platform

Large-Scale Ed-Tech Ecosystem

- Designed **comprehensive education platform** with React frontend, Go backend, and microservices architecture covering student/faculty management, course delivery, and assessment systems.
- Built credential verification system with DID generation, wallet management, and certificate issuance using Repository-Service pattern and event-driven architecture.
- Developed educational search engine with personalized ranking algorithms, context-aware suggestions, and Redis-backed caching for optimized query performance.

TECHNICAL SKILLS

Expert: Go, TypeScript, SQL, JavaScript

Proficient: C++, Python, Java, JSON-RPC

Familiar: Rust, Julia, Solidity, GraphQL, Server-Sent Events (SSE)

AI & LLM Infrastructure: Model Context Protocol (MCP), AI Agents, Vector Search, RAG Pipelines

Frontend: React, Material UI, Redux, HTML5, CSS3, Tailwind CSS, Nx Module Federation

Backend: Gin, Node.js, Express, REST APIs, JSON-RPC, Server-Sent Events (SSE)

Databases & Caching: Redis, Google Cloud Datastore, MySQL, PostgreSQL, MongoDB

Cloud & DevOps: Google Cloud Platform (GCP), Amazon Web Services (AWS), Docker, Kubernetes, CI/CD

Architecture: Microservices, Repository-Service Pattern, Mediator Pattern, Event-Driven Systems

ACHIEVEMENTS

- Open Source Contributor

Maintained public Go SDK with 100+ GitHub stars for IoT device management
- LeetCode

Solved 424+ problems with contest rating 1540, demonstrating strong DSA fundamentals and problem-solving consistency

EDUCATION

- PDPM IIITDM Jabalpur, India

(Jul 2018 - Aug 2022)

Bachelor of Technology in Mechanical Engineering

(GPA: 7.8/10.0)

Relevant Coursework:

Data Structures, Algorithms, Operating Systems, Database Systems, Computer Networks